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Compact FIN defected ground structure for improved cross polarisation in array antenna

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ABSTRACT

In this article, a 2 × 2 microstrip array antenna integrated with FIN defected ground structure (DGS) is investigated, which is designed for 3.24 GHz band operation. The FIN-DGS has been etched from the ground plane in the H-plane direction between two contiguous elements in a symmetric manner. Therefore, it suppresses the cross-polarisation (XP) radiation in the antenna without affecting its co-polarisation (CP) radiation characteristics. A proposed antenna has been designed and experimentally verified and shows <35 dB XP (±50°) radiation around boresight of the radiating elements in the H-plane with 10.7 dBi gain. The variation of XP fields has been investigated using the modal field distribution. The designed antenna provides a wide range of applications in adaptive arrays and space antennas, where 3D coverage with decreased XP values is the primary constraint.

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KEYWORDS
Polarisation; defected ground structure; microstrip antenna; array

1. Introduction

In the recent era, the level of cross-polarisation (XP) in the microstrip antenna array is the principal component for communication systems. The microstrip patch antenna resonates at TM_{10} (fundamental mode) and it radiates linearly in E-plane. However, some amount of orthogonal fields (TM_{02}) in H-plane is referred as XP radiation patterns (Huynh et al., 1988). The level of XP has been considered around 15–20 dB in H-plane below the peak gain. Maintain the XP radiation level in the array antenna while frequency reuse in the communication system has major limitations. Therefore, defected ground structures have been considered as a promising candidate in the microstrip array antenna (Petosa et al., 1999)–(Guha & Antar, 2010).

Several research articles have been presented to improve the XP radiation pattern and return loss bandwidth, and port isolation level is demonstrated by basic DGS methodology (Gu et al., 2017; (Monica et al., 2020, Niu et al., 2019; Pasha et.al., 2018; Zeng et al., 2018;). In (Gosh et al., 2017), a cross-type defected ground structure integrated microstrip patch antenna for wide bandwidth and improved CP-XP radiation. In (Pawar et al., 2019), a quasi planar composite microstrip antenna consist of a single layer rectangular microstrip antenna along with two metallic post was investigated for high gain with reduced CP-XP radiation.

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In (Zonunwaii, 2021), a pair of shorting strips are placed in the H-plane of circular patch to reduce surface issues and increase the gain with reduced XP radiation. A planar air-gap-loaded rectangular microstrip antenna integrated with single shorting post and defected ground structure was investigated by Sarkar et al. (2020). The above mentioned works has limitations of cost and fabrication complexity. Therefore, researchers have focused their efforts on improving the parameters of the microstrip array antenna based on the DGS structure. In (Pawar et al., 2019), a bracketed stub-loaded microstrip array antenna was investigated for excellent impedance matching and to reduce spurious radiation.

In (Susamay et al., 2018), the theoretical studies of circular microstrip patch with clusters of shorting pin placed symmetrically were investigated to show the reduction of XP in the H-plane. The observations of the above work have no significant enhancement in CP radiation. To improve the CP and reduce the XP radiation pattern level, a pair of dot DGS was used in the circular patch results in 5 dB reduction (Guha et al., 2005). The limitation of dot DGS has minimal effect on XP radiation that leads to an augmentation of an arc and annular ring DGS provides a subsequent suppression of 10–12 dB (Kumar & Guha, 2012). Apart from a circular patch, a rectangular patch with slot type defect was explored with suppression of XP about 15–25 dB. As the defect width of the slot increases, the XP fields were suppressed without affecting its dominant mode, but it limits in utilising large array (Gosh et al., 2015).

In the above mentioned works, DGS was deployed symmetrically near the non-radiating region in microstrip patch antenna to weaken the higher-order modes. This imposes a limit to the maximum size of geometries, which is overcome by resonant DGS and it is non-restrictive in size, where the size of DGS is more than patch geometry. Although, investigation inherits a new geometry to improve the XP in microstrip array that is asymmetric DGS (combination of both dominant and higher-order modes; Kumar & Guha, 2016; Zulkifli et al., 2008). In (Xiao et al., 2011), the DGS was used to improve the XP and suppress mutual coupling in the microstrip arrays antenna. In (Al-Dulaimi et al., 2020), a meander monopole antenna array with Hilbert-shaped EBG structure to reduce coupling effect between the antenna element. In (Ahmed & Elwi, 2019), a mutual coupling reduction between two monopole antenna element provides a bore-sight gain about 2dBi at 2.45 GHz.

To address the XP issue in microstrip array antenna, a flexible shaped DGS was also introduced to improve the CP to XP radiation (Kumar et al., 2017). In the other approach, asymmetric and compact configuration DGS were investigated to maintain the highest order of achievable suppression of cross-polarised (XP) radiations (Kumar & Guha, 2020). For high gain, improved XP and reduction in mutual coupling in the microstrip array antenna using DGS was investigated in (Zhu et al., 2020). Anyhow, the related research articles show how to improve the XP in microstrip antenna array using DGS without damaging the performance of the antenna array. The present study explores DGS to improve the stability of the pattern and the XP radiations.

Here, the authors investigated 2 $\times$ 2 microstrip array antenna with FIN-DGS at 3.24 GHz (lower frequency) for S-band applications. This design is significantly focused on the gain enhancement and reduction of XP radiation patterns. The FIN-DGS is etched from the ground plane in the H-plane direction between two contiguous elements in a symmetric manner. It follows the radiating element contour path to suppress the fields near the non-resonating region. It does not interrupt the primary radiating region so that

the dominant mode is not affected. The measured XP radiation pattern level of < 35 dB ($\pm 50^{\circ}$) is achieved with a maximum gain of 10.7 dBi and efficiency 87%. The prototype of the proposed array antenna is designed, fabricated and experimentally verified.

2. Antenna configuration

2.1. Antenna geometry

The configuration of a proposed 2 $\times$ 2 microstrip array antenna with the symmetric pair of FIN-DGS is shown in Figure 1 with optimised dimensions. The proposed antenna is investigated in S-band (3.24 GHz) employing in RT/Duroid 5880 substrate with ε_r =2.2, tan δ =0.0012 and thickness of 1.575 mm. The radiating elements ($L_p \times W_p$) are printed on top of the substrate. The distance between the radiating elements (d) from the centre is 0.5 $\lambda_\circ$ ($\lambda_\circ$ =resonant wavelength in free space). The size of the ground plane is 1.5$\lambda_\circ \times$ 1.5$\lambda_\circ$ (L $\times$ W), located on the backside of the substrate. The FIN-DGSs are etched from the ground in H- plane with a proper interleaving gap (g) on each side of the arm. As a result, the contour path of the radiating elements are followed by the FIN-DGS. In the proposed model, three DGS units are used for a pair of patches.

2.2. Investigation using single element

Investigation of FIN-DGS integrates 2 $\times$ 2 array antenna is planned with a standalone patch (Balanis, 2011). To validate the single element geometry the ratio of W_p/L_p = 1 is chosen with folded-in and folded-out FIN-DGS. The input impedance of the square microstrip patch varies from minimum to maximum values by moving the feed point

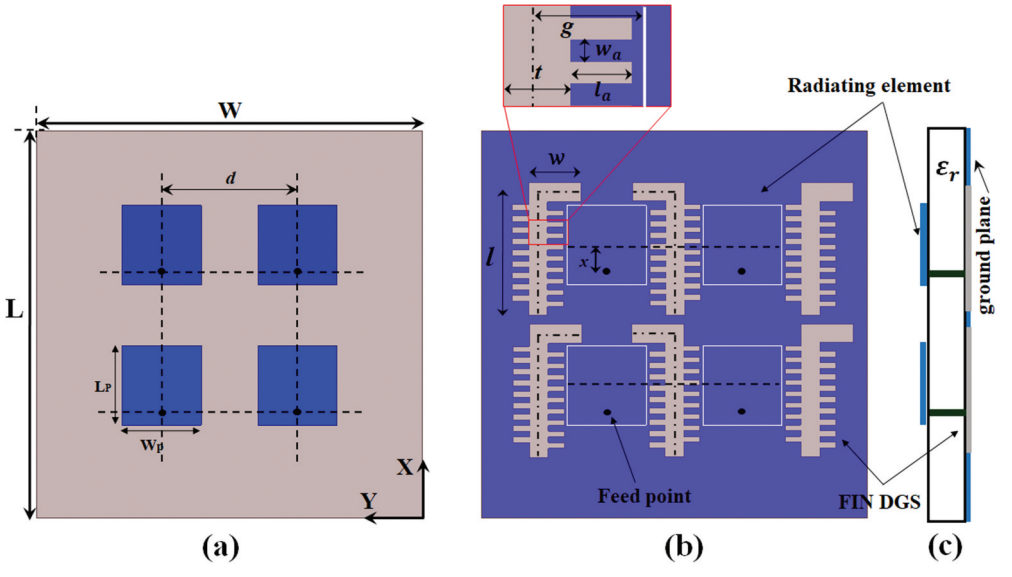


Figure 1. Geometry of the proposed array antenna model with FIN- DGS: L_g =135 mm, W_g =135 mm, L_p =28 mm, W_p =28 mm, x =5.5 mm, l =35 mm, w =14 mm, l_a =2 mm, w_a =1 mm, g =2 mm, t =3 mm, d =45.45 mm.

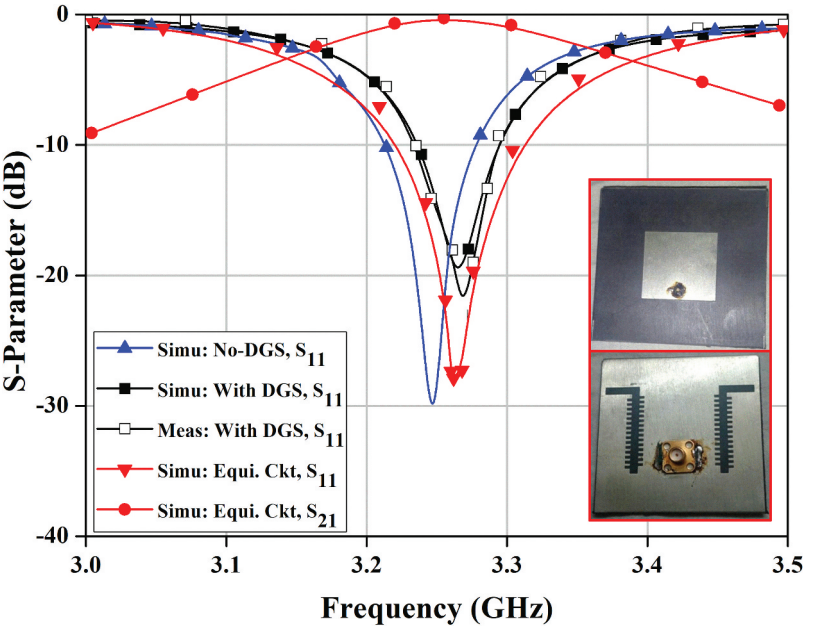


Figure 2. S-parameters and fabricated prototype of FIN-DGS with single element antenna.

from the centre towards the edge. Therefore, the feed point is not located at the centre of the patch; it is at 50 Ω impedance matching point. The location of the feed point (x) is 0.061 $\lambda_\circ$ from the centre of the radiating element. As a result, the electrical length increases and the resonant frequency shift to the lower frequency at 3.24 GHz. The reflection coefficient (Γ) of the proposed geometry is calculated using Equation (1).

$$\Gamma = \left|\frac{(Z_0 - Z_{in})}{(Z_0 + Z_{in})}\right| \tag{1}$$

where, Z_0 and Z_{in} denotes the characteristics and input impedance, respectively. The optimised parameters of FIN-DGS: l = 35 mm, w = 14 mm, g = 2 mm, t = 3 mm are taken to validate the geometry and the performance is shown in Figure 2. With a single element fabricated prototype in the inset. It is observed that the single element antenna is resonating at 3.24 GHz with an attenuation level of −28 dB. There is a minimal marginal shift in frequency may be due to inductive loading. Overall, the proposed S-parameters analysis shows good agreement with the EM simulation and measured results.

2.3. Modelling of FIN DGS

The equivalent circuit model of FIN-DGS and 3D structure is shown in Figure 3. The FIN-DGS is etched from the ground plane of the proposed geometry. Therefore, the defects on the ground plane disturb the current distribution; this disturbance changes the characteristics of a transmission line. The intrinsic series inductance (L) and shunt capacitance (C) of the ideal transmission line also changes and coupled with the defected ground plane

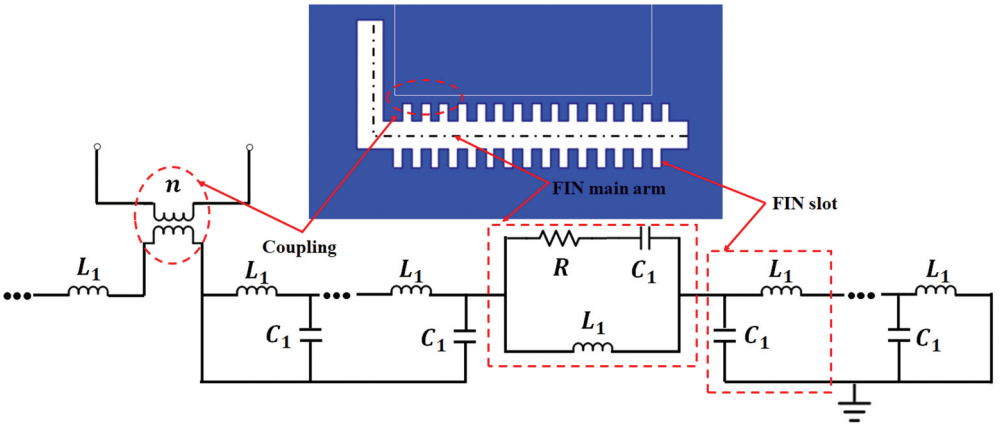


Figure 3. Equivalent circuit of single element with FIN-DGS.

structure. An ideal transformer is used for modelling the coupling between the microstrip line and the slots embedded on the ground plane. The ideal transformer turns ratio n is given by Caloz et al. (2004)

$$n = \sqrt{\frac{Z_{in}}{Z_L}} \cong \sqrt{\frac{Z_0^{\mu strip}}{Z_0^{slot}}} \tag{2}$$

where, $Z_0^{\mu strip}$ and Z_0^{slot} are the characteristic impedances of microstrip line and the slot, respectively. A lossy characteristic of the coupling between the microstrip line and FIN-DGS is considered to show the accuracy of the proposed DGS model. Since the equivalent circuit model of FIN-DGS is similar to the bandstop filters and operates at a particular resonant frequency. Therefore, the equivalent parameters of the circuit model are extracted from the physical structure. First, the input admittance of the slot (Y_{in}^{slot}) is calculated by using Equation (2):

$$Y_{in}^{slot} = \frac{1}{Z_0^{slot}\left[\frac{1-j\tanh \alpha l \cot \beta l}{\tanh \alpha l - j\cot \beta l}\right]} \tag{3}$$

where, α is the attenuation constant, β is the phase constant, and l is the electrical resonant length. Here, the input admittance of the slot (Y_{in}^{slot}) can be viewed loading of microstrip line (Y_L), and depends on transformer turn ratio n. Therefore, the input admittance of microstrip line to DGS, $Y_{in}^{\mu l-slot}$, can be determined by:

$$Y_{in}^{\mu l-slot} = \frac{Y_L}{n^2} = \frac{1}{R_0} + jB_0 \tag{4}$$

where, R_0 is the resistance of the equivalent circuit model, corresponded with a lossy characteristics within the attenuation pole frequencies of the coupling of microstrip-to-DGS. Hence, resistance (R_0), capacitance (C_0) and inductance (L_0) of the equivalent circuit are given by Wu et al. (2008)

$$R_0 = \left[\Re\mathfrak{e}(\mathfrak{Y}_{\mathrm{in}}^{\mu l-\mathrm{slot}})\right]^{-1} \tag{5}$$

$$C_0 = \frac{\Im m\left[\frac{\mathfrak{Y}_{in}^{\mu l-slot}}{n^2}\right]}{2\pi f_0\left[\frac{f_0}{f_c} - \frac{f_c}{f_0}\right]} \tag{6}$$

$$L_0 = \frac{1}{4\pi^2 f_0^2 C} \tag{7}$$

where f_0 is the attenuation frequency and f_c is the cut-off frequency of the DGS. The extracted equivalent circuit parameters are optimised for the proposed model, as listed in Table 1. The extracted circuit parameters are used in the circuit simulation tool to verify the equivalent circuit model. The proposed equivalent circuit analysis has a good agreement with the EM simulation and measurement.

Next, the proposed FIN-DGS with a microstrip radiating element is investigated using the E-field surface distribution as shown in Figure 4. In case of No-DGS, the electric fields are distributed along X-axis due to dominant mode near the radiating side. Moreover, the weak undesired fields are distributed along y-axis due to higher order mode near non-radiating side (H-plane). Hence, the higher intensification of undesired fields along H-plane is distributed in asymmetric manner, results in reduction in XP radiation. It is also observed in the proposed geometry, the high intensity electric field that existed in H-plane are asymmetric in nature and strongly inclined near patch centre. When the DGS is introduced in H-plane, the fields get coupled and resonate around desired frequency. Therefore, the distribution of fields are symmetric manner and it suppresses XP radiation. In case of FIN-DGS (folded-in and folded-out) which is placed in H-plane, the electric fields gets interact and coupled with radiating element. Therefore, the intensification of electric field decreases in H-plane results suppresses XP radiation. Next section provides the suppresses XP radiation mechanism of the designed antenna with FIN-DGS for better understanding.

2.4. XP radiation performance

There is always a XP component present whenever the antenna has a linear polarisation component. This is due to the gap field, which wraps around the patch, and thus exists in both coordinate directions. Based on Ludwig's first definition of XP and near-field approach (Ludwig, 1973), XP field level has been calculated in the square patch. Later, Ludwig's first equation is replaced by the Ludwig third equation (Hansen, 1987) because the polarisation of the radiated fields by an antenna varies in the far-field region. Hence, the observation angle is moving away from the broadside direction. Therefore, the XP radiation component (E_{xp}^{ϕ}) in the far-field region for the patch antenna can be written as (Bhardwaj & Rahmat-Samii, 2014)

$$E_{xp}^{\phi} = E_{\theta}\cos(\phi) - E_{\phi}\sin(\phi) \tag{8}$$

Table 1. Extracted values of circuit parameters of FIN DGS.

Parameters	L_1 (nH)	L_2 (nH)	C_1 (pF)	C_2 (pF)	R (Ω)	N
Values	0.0208	1.189	0.12	1.958	0.3	0.73

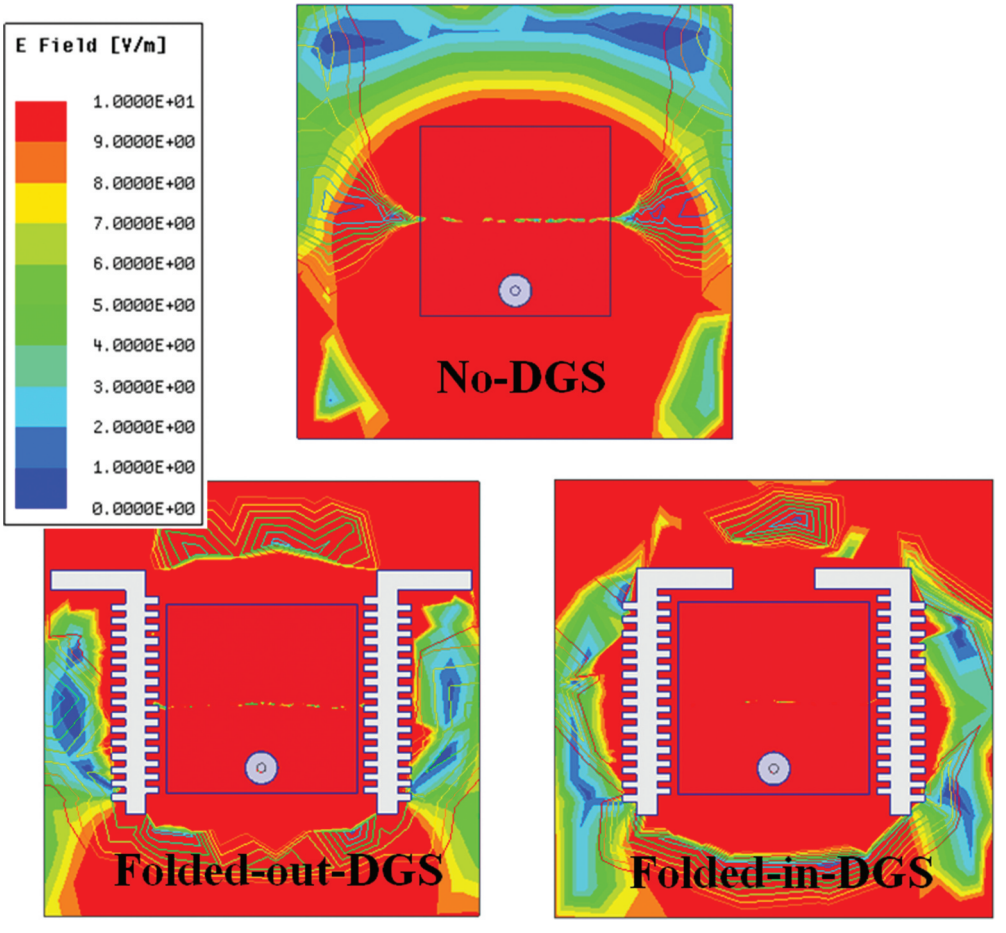


Figure 4. Surface current distribution of single element without and with FIN-DGS.

where E_θ and E_ϕ are the far-field electric field in azimuth and elevation E-plane component.

Equation (8) depicts the relationship between the patch antenna's near field and the far-field XP in the region of interest. As a result, the level of XP is defined as the maximum cross-polarisation ($E_{xp}^{dB}(\theta,\phi)$) in the broadside relative to the maximum co-polarisation ($E_{co}^{dB}(\theta,\phi)$) in the far-field H-plane and written as:

$$XP = \max\left[E_{xp}^{dB}(\theta,\phi)\right]_{0<\theta<\pi} - \max\left[E_{co}^{dB}(\theta,\phi)\right]_{0<\theta<\pi} \tag{9}$$

The 2D simulated CP and XP radiation pattern of the single element patch antenna with FIN-DGS is shown in Figure 5. It is observed that the maximum simulated and measured XP level is < 34 dB at 3.24 GHz obtained from Equation (9). The measured results closely follow the simulation predictions and confirm a reduction in XP radiation patterns. Here, we are only presenting the folded-out FIN-DGS results because it is almost equal to the folded-in FIN-DGS.

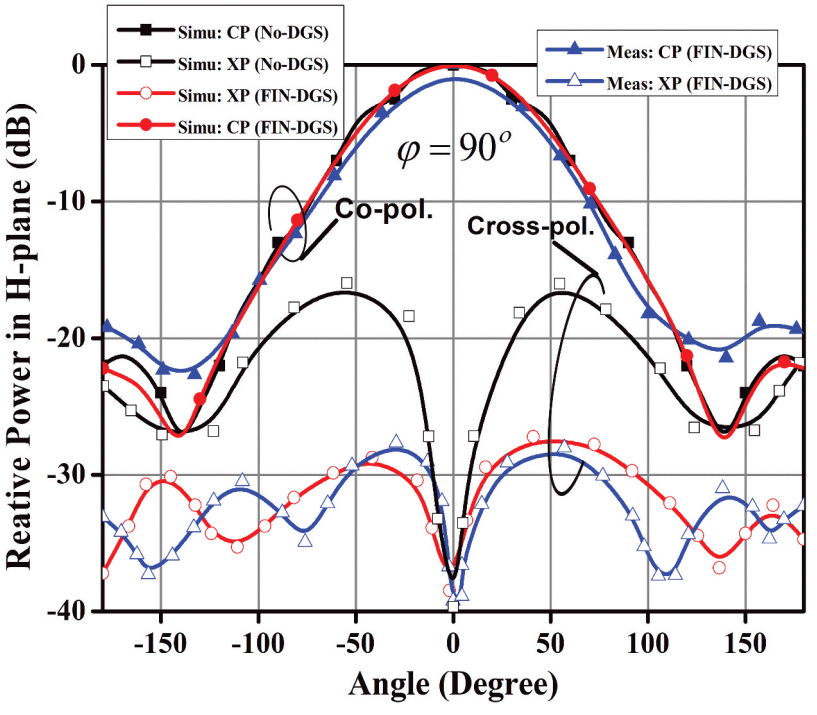


Figure 5. Simulated and measured H-plane radiation pattern of single element FIN-DGS.

3. Antenna array results and discussions

The fabricated prototype of a microstrip 2 × 2 array antenna integrated with FIN defected ground structure (DGS) is shown in Figure 6(a,b). The radiating elements and ground plane integrated with FIN-DGS are printed on the top and bottom of the substrate, respectively.

The distance between the radiating elements from the centre is taken as $d = 45.45$ mm. In the case of DGS, the gap between the folded-out and folded-in FIN-DGS are obtained by parametric study as shown in Figure 6(c). Here, we can observe that the maximum suppressions of XP are obtained by varying the defect at various distance of d_1, d_2. The optimised value is taken as $d_1 = 41.23$ mm and $d_2 = 17.54$ mm, respectively. The d_1, d_2 are used to control the ground plane current and it also reduces the mutual coupling. To validate the prototype, S-parameters and radiation patterns were measured using a vector network analyser and an anechoic chamber, respectively, both are available at Antenna Design Lab, VIT University, Vellore, India. The antenna is placed with the help of wooden support in an anechoic chamber. It is fed by identical SMA connectors.

The simulated and measured S-parameters of the proposed array antenna with and without FIN-DGS are shown in Figure 7. The S-parameters were measured at all the ports and only one port result is shown for comparison for the sake of brevity. The proposed microstrip 2 × 2 array antenna is resonating at a lower frequency of operation around 3.24 GHz. The simulated and measured S-parameters are found to be well matched. Aside from that, only a minor shift in frequency to the lower side is perceived. This could be due to capacitive loading, FIN-DGS fabrication tolerance, SMA connector soldering, or any

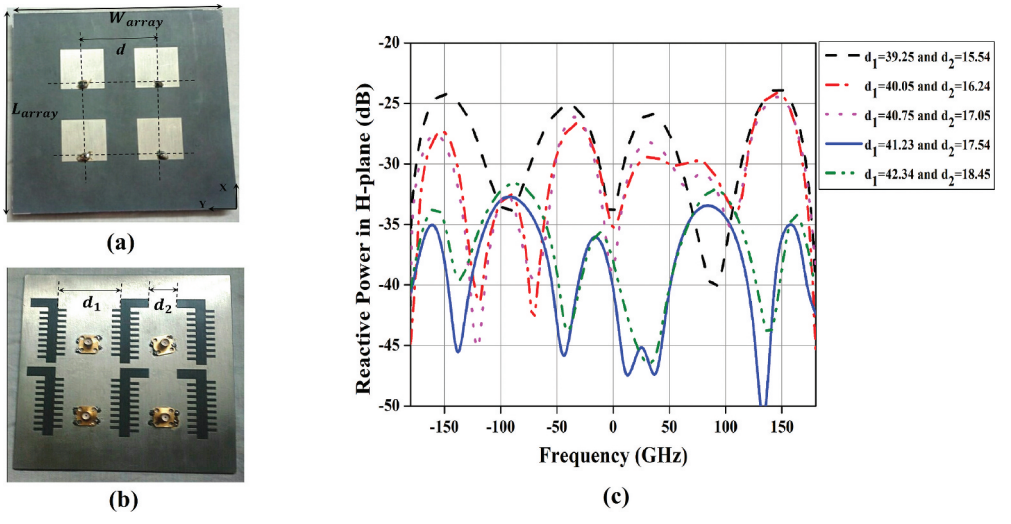


Figure 6. Microstrip 2 × 2 array antenna integrated with FIN-DGS; (a) photograph from patch side (b) photograph from ground plane side; (c) Parametric study various d_1 and d_2.

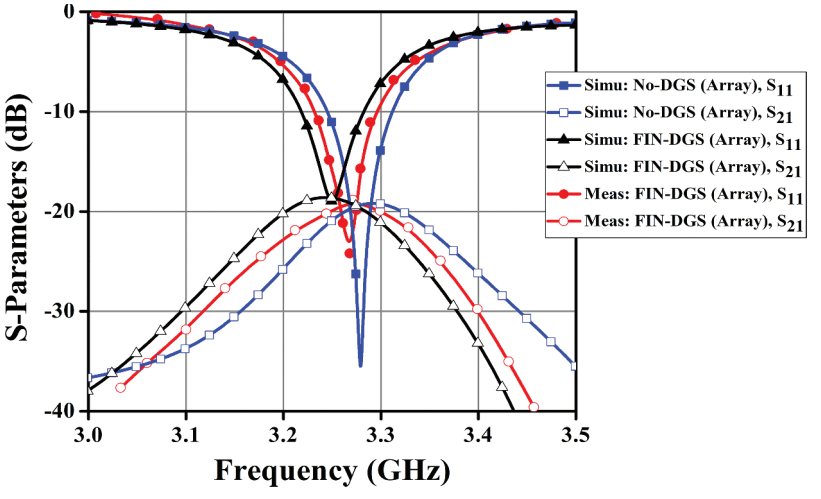


Figure 7. S-parameters of microstrip 2 × 2 array antenna integrated with FIN-DGS.

other external influence on the input impedance. Figure 7 also shows that the measured isolation of *S*21 between the antennas is $-20dB$ at 3.24 GHz. The isolation plot between the port is quite significant, revealing no change in the coupling even in the presence of the DGS. This signifies that the DGS in the proposed design interacts with the higher

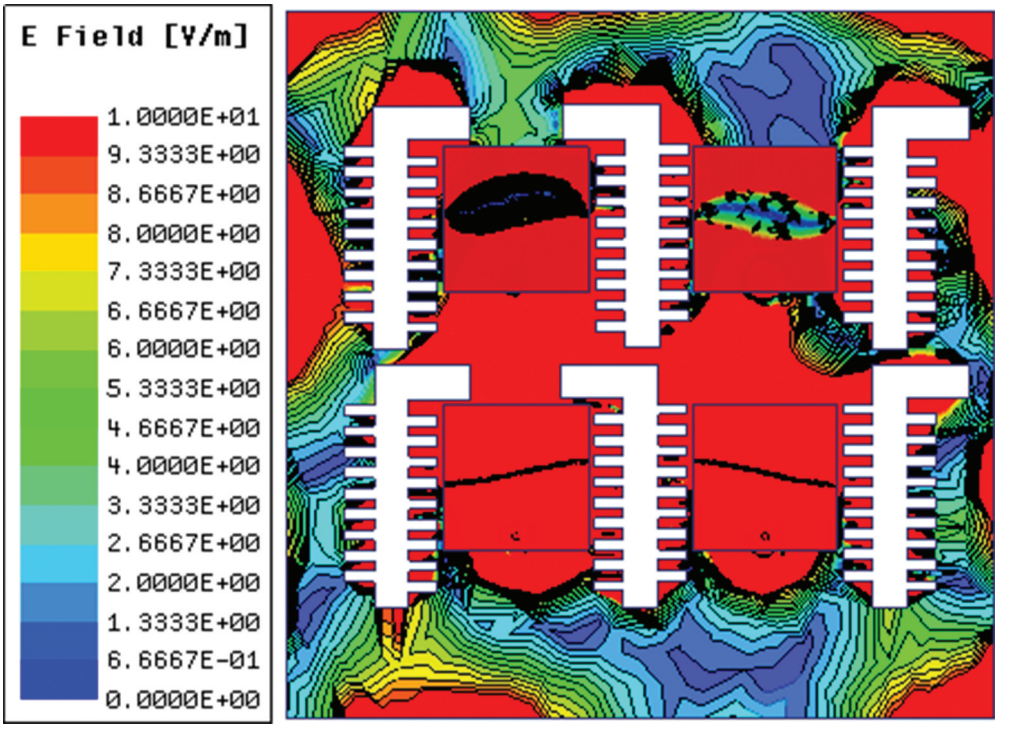


Figure 8. Surface current distribution of microstrip 2 × 2 array antenna integrated with FIN-DGS.

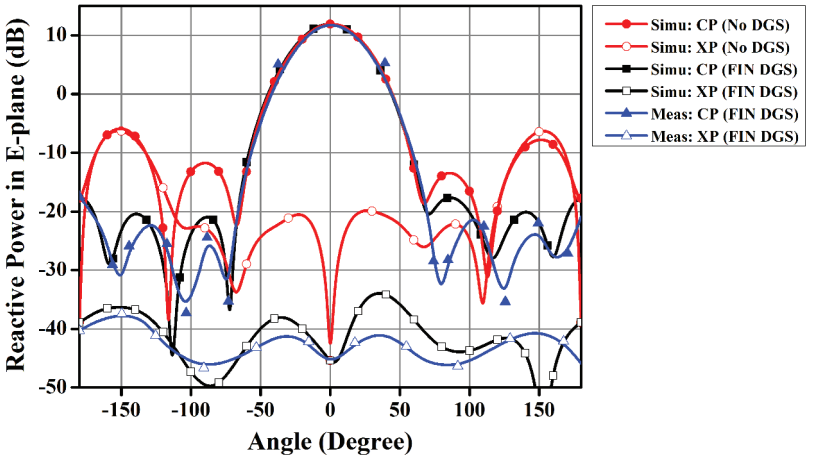


Figure 9. Simulated and measured radiation characteristics of microstrip 2 × 2 array antenna integrated with FIN-DGS at 3.24 GHz in E-plane ($\phi = 0°$).

mode but not with the surface wave. Figure 8 shows the surface current distribution of 2 × 2 array antenna with FIN-DGS, the electric field in H-plane gets coupled with FIN-DGS results the symmetric field distribution and reduced XP radiation.

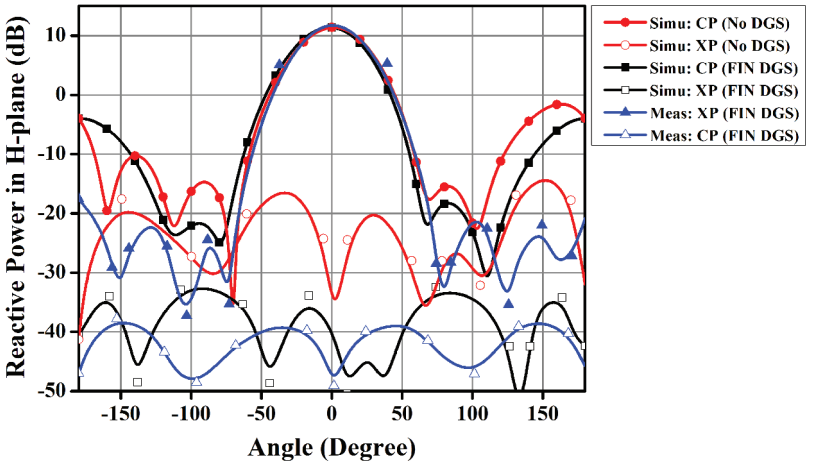


Figure 10. Simulated and measured radiation characteristics of microstrip 2 × 2 array antenna integrated with FIN-DGS at 3.24 GHz in H-plane ($\phi = 90°$).

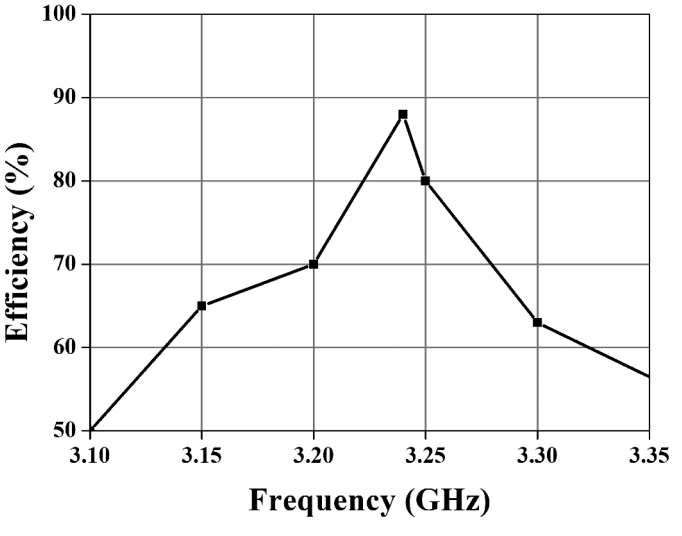


Figure 11. Efficiency (%) of the microstrip 2 × 2 array antenna integrated with FIN-DGS.

Figures 9 and 10 show the simulated and measured radiation patterns of the proposed 2 × 2 array antenna in E-plane ($\phi = 0°$) and H-plane ($\phi = 90°$) for No-DGS and FIN-DGS at 3.24 GHz. It is observed that, no considerable effect of FIN-DGS on E-plane CP radiation, whereas < 35 dB of XP level is observed in simulation and measured data. Therefore, we refrain from E-plane radiation in the present paper and our discussions are mainly restricted to H-plane XP for the sake of brevity. The simulated and measured radiation patterns are examined in H-plane, as shown in Figure 10. It is found that < 35 dB ($\pm 50°$) of XP level is achieved with good agreement between the simulated and measured radiation patterns. Finally, the results obtained are more than 35 dB XP isolation in both

Table 2. Comparison between the proposed work and recent literature.

Ref.	Frequency (GHz)	Antenna geometry	DGS shape	Ground plane size	XP level (dB)	CO-XP level in angular region	Gain (dBi)
Zeng and Luk (2018)	1.38–3.5	Dipole (Single element)	Slot in patch	1.44 λ_0	15	–	7.2
Niu et al. (2019)	4	Rectangular patch (1 × 3 array)	T and Ring type	0.59λ_0 × 1.02λ_0	–	–	3.82
Gosh et al. (2017)	4.91	Rectangular patch (Single element)	Cross type	1.34λ_0 × 1.34λ_0	–	25–28 dB (±45°)	6–7
Pasha et al. (2018)	6	Rectangular patch (Single element)	Composite type	λ_0 × λ_0	10–14	5 dB (±45°)	8.4
Zonunwaii (2019)	9.16	Circular patch (Single element)	shorting strips stub	1.84λ_0 × 1.84λ_0	–	25 dB (±70°)	10
Sarkar et al. (2020)	8.3–15.20	Rectangular patch (Single element)	Slot DGS	1.66λ_0 × 1.66λ_0	–	16.5 dB (±70°)	5–6
Pawar (2019b)	10.22	Rectangular patch (2 × 1 array)	Bracketed stub	2λ_0 × 2.5λ_0	25–35	27 dB (±180°)	10
Susamay et al. (2018)	3.5	Circular patch (single element)	Cluster of shorting pins	0.7λ_0 × 0.7λ_0	51	35 dB (±180°)	7.4
Al-Dulaimi et al. (2020)	2.45	Monopole antenna (1 × 2 array)	Hilbert Shaped	0.33λ_0 × 0.41λ_0	–	–	2.1
Ahmed and Elwi (2019)	2.45	Monopole antenna (1 × 2 array)	Minkowski like pre fractal shaped	0.41λ_0 × 0.20λ_0	–	–	2
Kumar and Guha (2012)	5.9	Circular patch (single element)	Angular ring	λ_0 × λ_0	25	11 dB (±60°)	–
Gosh et al. (2015)	8.2	Rectangular patch (single element)	Slot type	3λ_0 × 3λ_0	27	25 dB (±50°)	–
Kumar and Guha (2016)	10	Rectangular patch (single element)	Folded-out type	2λ_0 × 2λ_0	25	15 dB (±50°)	6.5
Kumar et al. (2017)	10	Rectangular patch (2 × 2 array)	Flexible type	2λ_0 × 2λ_0	25	12 dB (±35°)	12.2
Zhu et al. (2020)	24.37–28.5	Vivaldi antenna (1 × 8 array)	Line type	0.55λ_0 × 0.55λ_0	24–26	15 dB (±90°)	7.7–8.9
This work	3.24	Square patch (2 × 2 array)	FIN type	1.5λ_0 × 1.5λ_0	35–40	35 dB (±50°)	10.7

planes concerning the co-polar peak. The effect of the array is evident as the beam width is reduced, resulting in the gain that is observed around 10.7 dBi, compared to 6 dBi for single element for CP radiation shown in Figures 9 and 10. Figure 11 shows the radiation efficiency of the proposed antenna. At 3.24 GHz the radiation efficiency is around 87% observed. The performance of the proposed antenna is compared with recent literature shown in Table 2. Here, the proposed structure exhibits good impedance matching along and polarisation purity with high gain, it is suitable in an array configuration. Therefore,

FIN-DGS integrated with square microstrip patch antenna has excellent CP-XP isolation in the entire elevation range may be helpful for the antenna community to meet modern wireless communication.

4. Conclusion

This paper describes a square microstrip 2 × 2 array antenna with FIN-DGS that significantly improves the XP radiation pattern. The FIN-DGS is positioned in the H-plane to effectively suppress the XP radiation pattern. It is simple to deploy in large arrays by halving the number of DGSs without sacrificing performance. The designed antenna has < 35 dB ($\pm 50^{\circ}$) of XP level with good agreement between the simulated and measured radiation patterns and a maximum gain of 10.7 dBi.It can improve the CP-XP ratio over a wide angular range, especially in wireless base-station networks, linear arrays, and other applications where high XP is a severe limitation. This should be excellent in practice and industry because it requires no additional space, weight, or cost, while also allowing for the development of other new antenna features in the near future.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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